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Efficient Switchback Experiments with Surrogate Variables: Estimation and Experimental Design

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Abstract. Switchback experiments, in which experimental units are assigned to control and treatment interchangeably over time, have gained increasing popularity in recent years due to their ability to mitigate interference across units. Despite their wide adoption in the technology sector, drawing reliable inferences from such experiments remains challenging due to temporal interference. In this paper, we address this challenge by examining a special causal structure commonly found in a wide range of applications. Specifically, we consider scenarios where there exist surrogate variables at each experimental period that can fully capture the potential temporal interference from the past. We outline the necessary assumptions for identification in this context and propose an unbiased estimator using the technique of importance sampling. Additionally, we derive an experimental design with a near-optimal worst-case guarantee and compare it theoretically and empirically to the inverse propensity score estimator. To facilitate inferences, we introduce both an exact inference procedure based on Fisher's randomization test and an asymptotic inference procedure based on the central limit theorem. We demonstrate by extensive numerical experiments that the proposed estimator, coupled with the experimental design, results in lower risk compared with the traditional inverse propensity score estimator.

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1. Introduction

Experimentation has become increasingly important in contemporary business operations due to its ability to deliver statistically rigorous insights of the impact of new policies, even under constrained budgets (Bakshy et al. 2014, Kohavi and Thomke 2017, Kohavi et al. 2020, Thomke 2020). However, the scale and complexity of modern digital platforms introduce new statistical challenges in the design and analysis of experiments (Kohavi et al. 2009, Chamandy 2016). A key challenge is the issue of interference between treatment units, where the outcomes for individual units are not solely determined by their assigned policies but are also influenced by the policies applied to other units. For example, consider a ride-hailing platform testing a new customer fare policy in a city. Traditional A/B testing, which randomly assigns half of the city districts to the old fare (control group) and the other half to the new fare (treatment group), would fail to provide an unbiased estimate of the treatment effect. This is because customers in both groups interact with the same pool of drivers, allowing the group with the fare advantage to monopolize the available supply, thereby introducing bias into the results.

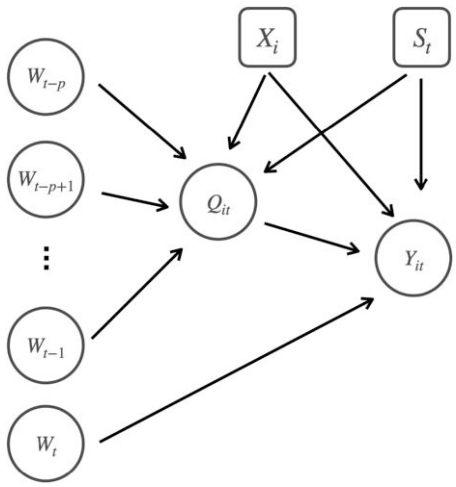
To address interference between treatment units, a commonly used strategy is to implement *switchback experiments* (Oman and Seiden 1988, Bojinov et al. 2023), in which the entire population is alternately assigned to the treatment or control group over time. This aggregation reduces cross-unit interference, making switchback experiments particularly popular in contexts such as online marketplaces or matching platforms, where interference issues are pronounced. However, although this approach mitigates cross-unit interference, it introduces a new type of interference known as the *carryover effect* (Bojinov and Shephard 2019). Carryover effects arise when past treatment assignments influence future outcomes. For instance, in the ride-hailing platform example, a customer's dissatisfaction with a previously assigned price might discourage them from engaging

with the platform in the future, thereby affecting the outcomes of subsequent experiments. Addressing this temporal interference is a critical challenge in the design and analysis of switchback experiments.

To mitigate carryover effects, the existing literature has proposed two primary approaches. The first is a parametric strategy that assumes a specific outcome model (Armitage and Hills 1982, Xiong et al. 2019, Abadie and Zhao 2021). The second adopts a fully nonparametric perspective based on the potential outcomes framework (Bojinov et al. 2023, Ni et al. 2023, Xiong et al. 2023). Parametric approaches are more efficient when the model is correctly specified but are susceptible to errors from model misspecification. In contrast, nonparametric approaches are more robust to outcome model assumptions but are less efficient in sample utilization. Without a well-defined structure for the carryover effect, data near switching points cannot be characterized effectively and are often discarded in standard estimators like the inverse propensity (IP) score estimator (Hu and Wager 2022, Bojinov et al. 2023). In this paper, we propose an intermediate approach that assumes a general outcome model with a structured representation of the carryover effect, commonly observed in many applications. This framework strikes a balance, achieving both sample efficiency and robustness to model misspecification.

We consider a scenario where an intermediate variable is accessible to the experimenter, capturing the carryover effect by blocking the causal path from past assignments to the current outcome. This variable serves as a *surrogate* for the past assignments (Athey et al. 2019). Figure 1 illustrates the causal relationships among the treatments W_t, the intermediate outcome Q_{it}, and the final outcome Y_{it}, where the subscripts i and t denote unit i and time period t, respectively. The covariates X_i and S_t represent observable characteristics of the unit and time, respectively. For instance, in the context of a ride-hailing platform, unit i would correspond to the ith district, and the intermediate variable could represent the customer activeness level in that district during a given period. Once customers visit the platform and observe the price, their decision to take a ride depends only on the current assignment and not on prior ones.

Figure 1. Causal Relationships Considered in This Paper



Note. Here, W_t is the treatment assignment at time t, Q_{it} and Y_{it} are the intermediate outcome and final outcome for unit i at time t, and S_t and X_i are the covariates corresponding to time and individual unit, respectively.

The causal structure in Figure 1 extends beyond two-sided matching platforms like ride-hailing services. For example, in cross-over designs in clinical trials (Laird et al. 1992, Ratkowsky et al. 1992), patients are alternately assigned to treatment and control groups over time. The carryover effect arises because previous assignments may affect patients' health conditions, which subsequently influence outcomes in later treatment periods. In this context, the intermediate variable could be the patients' health condition at the beginning of each experimental period. Another example is a retailer testing a new promotion planning algorithm (Ferreira et al. 2016), where the intermediate variable could represent competitors' prices, as these continuously respond to previous promotions and influence customers' current behaviors.

This paper addresses two primary challenges under the described causal structure. The first is constructing a more efficient estimator for the treatment effect of a given policy. The second is designing experiments that leverage this structure to improve estimation. We tackle these questions by introducing a novel estimator based on importance sampling and devising an experimental design that minimizes the risk associated with this estimator.

1.1. Main Contributions

This paper explores how to leverage auxiliary variables in switchback experiments in a fully nonparametric framework. The main contributions are as follows. (1) We identify the common structure of surrogate variables in experimental settings and theoretically establish the corresponding identification assumptions, enabling the rigorous use of intermediate variable information. (2) We propose a novel estimator based on importance sampling that exploits the surrogacy structure. By incorporating the likelihood ratio of the intermediate variable at each time period, our estimator efficiently utilizes all data, including those from switching periods, which are typically discarded in traditional IP score estimators (Bojinov and Shephard 2019, Hu and Wager 2022). (3) We develop a simple yet near-optimal experimental design based on our estimator under the minimax framework. This design is demonstrated to reduce the risk of estimators compared with traditional naive design.

1.2. Related Work

This paper is related to two streams of literature: experimental design and the surrogacy model. In experimental design, traditional studies often focus on small-scale problems in controlled environments. For instance, Fisher (1936) pioneered experimental design in agriculture, and Box and Draper (1971) introduced the well-known factorial design for industrial manufacturing.

Advancements in technology have introduced new challenges due to the increased scale and complexity of experimental settings. Recent research has focused on addressing interference in experiments. Glynn et al. (2020) modeled the experimental process as a Markov chain, using the steady-state reward as the outcome. Similar approaches can be found in Farias et al. (2022), Johari et al. (2022), and Hu and Wager (2022). Another line of work models interference through network structures, assuming that a unit's treatment only affects its adjacent neighbors (Ugander et al. 2013, Sussman and Airoldi 2017, Athey et al. 2018, Puelz et al. 2019). Additional research focuses on marketplace-specific settings, modeling interference through market equilibrium dynamics (Wager and Xu 2021, Johari et al. 2022, Li et al. 2022).

There has been an interest in developing experiments specific to the temporal setting as well. The most common design is the switchback design (Hu and Wager 2022, Bojinov et al. 2023, Ni et al. 2023, Xiong et al. 2023), where treatment and controls are assigned interchangeably. Other designs include the staggered rollouts (Xiong et al. 2019) and synthetic controls (Abadie and Zhao 2021). Our model distinguishes from the aforementioned literature by incorporating an auxiliary variable that introduces structure to temporal interference.

This paper is also closely related to the literature on surrogacy in econometrics, which is used to handle the missing data problem when estimating long-term treatment effects (Prentice 1989, Begg and Leung 2000, Frangakis and Rubin 2002, Athey et al. 2018, Anderer et al. 2022). In these contexts, surrogate outcomes are postulated as intermediate variables resulting from past treatment and are used to predict unknown future outcomes. This paper addresses a different problem from the literature mentioned above. In contrast, this paper adapts the surrogacy framework to a longitudinal setting under switchback treatment assignments, focusing on utilizing surrogate variables for more efficient statistical inference. A key distinction from Athey et al. (2018) is that we do not assume a stationary environment; instead, we require only the surrogate to be stationary, enabling more flexible estimation and inference.

1.3. Roadmap

The remainder of the paper is structured as follows: Section 2 formulates the problem using the potential outcomes framework and outlines the identification assumptions. Section 3 introduces a novel estimator based on importance sampling. Section 4 focuses on experimental design, aiming to minimize the maximal risk of the proposed estimator. We frame this as a nonconvex optimization problem and propose a near-optimal heuristic solution. Section 5 discusses inference procedures, presenting both exact and asymptotic methods. Section 6 presents extensive numerical experiments to demonstrate the efficiency and robustness of our approach. Finally, Section 7 concludes the paper.

2. Problem Formulation

2.1. Setup

Consider an experimenter aiming to understand the treatment effect of a new policy by conducting a switchback experiment over T time periods. There are N experimental units denoted by $[N] = \{1, 2, \ldots, N\}$, each endowed with a pretreatment covariate $\boldsymbol{X}_i \in \mathbb{R}^d$ that encompasses individual characteristics of the unit. Under the setup of switchback experiments, at each time step $t \in [T] = \{1, 2, \ldots, T\}$, all the test units will be assigned to either treatment or control. In this way, we use $W_t \in \{0, 1\}$ to denote the random treatment assignment at time t with zero standing for control and one standing for treatment. For any two time steps $t_1 \le t_2$, the assignment path $\boldsymbol{W}_{t_1:t_2} = (W_{t_1}, W_{t_1+1}, \ldots, W_{t_2})$ denotes the collection of assignments from time t_1 to t_2. We adopt the convention that the upper case $\boldsymbol{W}_{t_1:t_2}$ stands for the random assignments and the lower case $\boldsymbol{w}_{t_1:t_2}$ stands for one realization. Finally, covariates $\boldsymbol{S}_t \in \mathbb{R}^l$ at each time t represent time-varying characteristics. Examples include weather at time t or the day of the week.

At each time t, the experimenter observes the outcome of interest $Y_{it} \in \mathbb{R}$ for each unit $i \in [N]$, which depends on the assignment path up to time t, that is, $\boldsymbol{W}_{1:t}$. The effect that previous assignments $\boldsymbol{W}_{1:t-1}$ has on the current outcome Y_{it} is called the *carryover effect* (Bojinov et al. 2023). Carryover effect makes it difficult to disentangle the effect of the current assignment from past assignments, which is the main challenge in estimating the treatment effect after conducting the switchback experiments (see, for example, Senn and Lambrou (1998), Bojinov and Shephard (2019), and Hu and Wager (2022) for more discussion).

In this paper, we leverage a common causal structure in many applications of switchback experiments. In addition to Y_{it}, we assume there is an intermediate outcome $Q_{it} \in \mathbb{R}$ for each unit i at time t that depends on the past assignments up to time $t-1$, that is, $\boldsymbol{W}_{1:t-1}$, which is also observable for the experimenter. We assume this intermediate outcome could capture the test units' response to the past assignment and serves as an indicator of the strength of the carryover effect. In the example of testing a new customer fare

policy in the ride-hailing platform stated in the Introduction, Q_{it} could be the number of customer visits at time t in the ith region.

We will formulate the problem in the framework of potential outcomes (Rubin 1974, Holland 1986). In particular, for each unit $i \in [N]$ at each time $t \in [T]$, we define a set of potential outcomes $\{Y_{it}(\boldsymbol{w}_{1:t})\}_{\boldsymbol{w}_{1:t}\in\{0,1\}^t}$ and $\{Q_{it}(\boldsymbol{w}_{1:t-1})\}_{\boldsymbol{w}_{1:t-1}\in\{0,1\}^{t-1}}$. These outcomes correspond to the hypothetical scenario where unit i follows a predefined treatment trajectory up to time t. For example, $Y_{it}(1_t)$ denotes the outcome for unit i at time t assuming it was assigned to treatment for all the t periods. Specifically, for a random assignment $\boldsymbol{W}_{1:T}$, each observed outcome is the specific potential outcome associated with the assignment, that is,

$$Y_{it} = Y_{it}(\boldsymbol{W}_{1:t}), \quad Q_{it} = Q_{it}(\boldsymbol{W}_{1:t-1}).$$

For notational convenience, we collect the potential outcomes into $\mathcal{Y}_i = \{Y_{it}(\boldsymbol{w}_{1:t})\}_{t\in[T],\boldsymbol{w}_{1:t}\in\{0,1\}^t}$ and $\mathcal{Q}_i = \{Q_{it}(\boldsymbol{w}_{1:t-1})\}_{t\in[T],\boldsymbol{w}_{1:t-1}\in\{0,1\}^{t-1}}$ as the aggregated potential outcomes for the i-th unit. In this paper, we postulate a superpopulation $\mathcal{F}$ that the covatiates and potential outcomes $(\boldsymbol{X}_i, \mathcal{Y}_i, \mathcal{Q}_i)$ for each unit i are independently sampled from $\mathcal{F}$.[1] As a result, there are two sources of randomness in our model. The first randomness originates from the potential outcomes itself, and the second one arises from the random assignment.

To make the model more tractable, we first assume the order carryover effect is finite.

Assumption 1 (p-Carryover Effect). *There exists a positive integer p such that for any $t > p$, if two assignment paths $\boldsymbol{w}_{1:t}$ and $\boldsymbol{w}'_{1:t}$ satisfy $\boldsymbol{w}_{t-p:t} = \boldsymbol{w}'_{t-p:t}$, we have $Y_{it}(\boldsymbol{w}_{1:t}) = Y_{it}(\boldsymbol{w}'_{1:t})$, and if $\boldsymbol{w}_{t-p:t-1} = \boldsymbol{w}'_{t-p:t-1}$, we have $Q_{it}(\boldsymbol{w}_{1:t-1}) = Q_{it}(\boldsymbol{w}'_{1:t-1})$.*

Assumption 1 states that the outcome at time t is independent of any treatment assignment preceding time $t-p$. This is a common assumption in the literature considering the carryover effect; for example, Bojinov et al. (2023) has the same assumption described here, and in the repeated measurement designs (Hedayat and Afsarinejad 1978), p is assumed to be one. It relies on the experimenter's domain knowledge to choose the right magnitude of p. A larger p will make the assumption generally looser and easier to satisfy in reality; however, it will lead to a more difficult estimation step as the long carryover effect will contaminate the outcomes. In the ride-hailing example, the effect of surge pricing on the platform typically dissipates after one or two hours as pointed out in Garg and Nazerzadeh (2022). Therefore, if the company sets the experimental interval to be half an hour, setting $p = 2$ would be an appropriate choice. With this assumption, we can simplify $Y_{it}(\boldsymbol{W}_{1:t})$ and $Q_{it}(\boldsymbol{W}_{1:t-1})$ to $Y_{it}(\boldsymbol{W}_{t-p:t})$ and $Q_{it}(\boldsymbol{W}_{t-p:t-1})$ for any assignments $\boldsymbol{W}_{1:t}$, which we will use for the remainder of the paper.

The objective of the switchback experiment is to estimate the average treatment effect of the new policy. In this paper, we will be focusing on the effect of the treatment if it were persistently assigned throughout the experimental horizon compared with its control counterpart. With this spirit, we define the following average treatment effect estimand τ:

$$\tau = \frac{1}{T-p}\sum_{t=p+1}^{T} \mathbb{E}[Y_{it}(\mathbf{1}_{p+1}) - Y_{it}(\mathbf{0}_{p+1})], \tag{1}$$

where $\mathbf{1}_{p+1}$ and $\mathbf{0}_{p+1}$ refer to the $p+1$-dimensional vector of all ones and zeros. This estimand captures the difference between implementing the new policy throughout the experiment and maintaining the control baseline and is widely accepted in the literature regarding the switchback experiments (Hu and Wager 2022, Bojinov et al. 2023). Note that, because we adopt the superpopulation perspective, the expectation here is taken with respect to the potential outcomes.

The fundamental challenge of estimating the treatment effect τ lies in the fact that we can never observe $Y_{it}(\mathbf{1}_{p+1})$ and $Y_{it}(\mathbf{0}_{p+1})$ simultaneously. Moreover, in the context of switchback experiments, the experimenter may observe neither $Y_{it}(\mathbf{1}_{p+1})$ and $Y_{it}(\mathbf{0}_{p+1})$ if the time t is around the switching time. In the next section, we introduce several identification assumptions that could allow us to do valid estimation on τ.

2.2. Identification Assumptions

We first focus on the problem of estimating the treatment effect τ given the observed outcome data. In the following section, we do not assume any specific experimental structure. Instead, we introduce several general identification assumptions regarding the treatment assignments and the potential outcomes. Any experimental mechanism satisfying these conditions is suitable for the estimation strategy proposed in the next section. To start with, we assume the treatment assignments and the potential outcomes are unconfounded.

Assumption 2 (Unconfoundedness). *We assume that, given covariates $\boldsymbol{S}_t$, the treatment assignments are independent of the potential outcomes, that is,*

$$\boldsymbol{W}_{1:T} \perp\!\!\!\perp (Y_{it}(\boldsymbol{w}_1), Q_{it}(\boldsymbol{w}_2)) | \boldsymbol{S}_1, \boldsymbol{S}_2, \ldots, \boldsymbol{S}_T \qquad \forall i \in [N],$$

for any $\boldsymbol{w}_1 \in \{0,1\}^{p+1}$ and $\boldsymbol{w}_2 \in \{0,1\}^p$.

Assumption 2 is common in causal inference on observational data (Rubin 1974, Imbens and Rubin 2015). It is more natural here because the experimenter can control the assignments $\boldsymbol{W}_{1:T}$. The second assumption concerns the role of the intermediate variable. Specifically, we assume that, conditioned on the variable Q_{it}, the outcome Y_{it} is independent of the

past assignments before $t-1$. In other words, Q_{it} blocks the causal path from $\boldsymbol{W}_{t-p:t-1}$ to Y_{it} and acts as a surrogate for the future outcome Y_{it}.

Assumption 3 (Surrogacy). *For each unit i, given the intermediate variable at the current time, the outcome is independent of the past assignments; that is, for any $t \in [T]$ and $i \in [N]$, we have*

$$Y_{it} \perp\!\!\!\perp \boldsymbol{W}_{t-p:t-1} | (Q_{it}, \boldsymbol{X}_i, \boldsymbol{S}_t).$$

Assumption 3 is analogous to assumptions in surrogate models in causal inference, such as assumption 2 in Athey et al. (2019). Intuitively, it implies that the intermediate variable fully captures the causal effect of earlier treatments on the current outcome. In other words, the past assignments affect the current outcome only through the intermediate variable. The causal relationship between treatments, the intermediate variable, and the outcome are illustrated in Figure 1.

Finally, to generalize results across time, we assume stationarity in the potential outcomes.

Assumption 4 (Stationarity). *Given the covariates, the potential outcomes are independent across time, that is,*

$$(Y_{it}(\boldsymbol{w}_1), Q_{it}(\boldsymbol{w}_2)) \perp\!\!\!\perp (Y_{it'}(\boldsymbol{w}_1), Q_{it'}(\boldsymbol{w}_2)) | \boldsymbol{X}_i, \boldsymbol{S}_1, \boldsymbol{S}_2, \ldots, \boldsymbol{S}_T,$$

for any $t \neq t'$, $\boldsymbol{w}_1 \in \{0,1\}^{p+1}$, and $\boldsymbol{w}_2 \in \{0,1\}^p$. Additionally, the distribution of the potential outcomes of the intermediate variable Q_{it} is the same across time and units when conditioned on the covariates; that is, for any $t, t' \in [T]$ and $i, i' \in [N]$, we have

$$Q_{it}(\boldsymbol{w}) | (\boldsymbol{X}_i = \boldsymbol{x}, \boldsymbol{S}_t = \boldsymbol{s}) \sim Q_{i't'}(\boldsymbol{w}) | (\boldsymbol{X}_{i'} = \boldsymbol{x}, \boldsymbol{S}_{t'} = \boldsymbol{s}),$$

for any assignments $\boldsymbol{w} \in \{0,1\}^p$ and covariates $\boldsymbol{x} \in \mathbb{R}^d$, $\boldsymbol{s} \in \mathbb{R}^l$.

Note that the independence assumption in Assumption 4 applies to potential outcomes and not observed outcomes because the latter is correlated through the treatment assignments. Furthermore, we do not assume stationarity in the potential outcomes of Y_{it}, relaxing the assumptions in surrogate models (Prentice 1989, Athey et al. 2019). The environment can be nonstationary over time as long as the intermediate variable is stationary. By choosing an appropriate surrogate variable and including suitable covariates $\boldsymbol{S}_t$ and $\boldsymbol{X}_i$, we are more likely to ensure stationarity.

Based on Assumptions 3 and 4, we define the surrogate index as the conditional expectation of the outcome given the treatment assignments and covariates.

Definition 1 (Surrogate Index). For any $t \in [T]$, $\boldsymbol{s} \in \mathbb{R}^l$, $\boldsymbol{x} \in \mathbb{R}^d$, $w \in \{0,1\}$, and $q \in \mathbb{R}$, the surrogate index is defined as the conditional mean of the outcome given the surrogate variable and covariates, that is,

$$h_t(q, w, \boldsymbol{s}, \boldsymbol{x}) = \mathbb{E}[Y_{it} | Q_{it} = q, W_t = w, \boldsymbol{S}_t = \boldsymbol{s}, \boldsymbol{X}_i = \boldsymbol{x}].$$

Additionally, the conditional intermediate outcomes at time t, given the previous treatment assignments and covariates, are denoted as

$$G(\boldsymbol{w}_{t-p:t-1}, \boldsymbol{s}, \boldsymbol{x}) \sim Q_{it} | (\boldsymbol{W}_{t-p:t-1} = \boldsymbol{w}_{t-p:t-1}, \boldsymbol{X}_i = \boldsymbol{x}, \boldsymbol{S}_t = \boldsymbol{s}).$$

The surrogate index can be estimated using regression because all inputs are directly observable. Serving as a bridge between past assignments and future outcomes, the surrogate index simplifies the carryover effect. By focusing on the effect of the intermediate variable on the outcome and the effect of past treatment assignments on the surrogate separately, we arrive at the following proposition.

Proposition 1. *Under Assumptions 2–4, the average treatment effect is given by*

$$\tau = \frac{1}{T-p} \sum_{t=p+1}^{T} \mathbb{E}[h_t(G(\mathbf{1}_p, \boldsymbol{S}_t, \boldsymbol{X}_i), 1, \boldsymbol{S}_t, \boldsymbol{X}_i) - h_t(G(\mathbf{0}_p, \boldsymbol{S}_t, \boldsymbol{X}_i), 0, \boldsymbol{S}_t, \boldsymbol{X}_i)].$$

Proposition 1 decomposes the treatment effect into two components. The first component, $G(\cdot,\cdot,\cdot)$, captures the impact of past treatment assignments on the intermediate variable. The second component, $h_t(\cdot,\cdot,\cdot)$, measures the current effect of the treatment given the intermediate outcome. This decomposition allows for efficient estimation by focusing on each component separately and will be used in devising the new estimator.

2.3. Discussion on Practical Relevance

Thus far, we have laid out the basic model and related assumptions in previous sections, which serve as the foundation for statistical inference. In this section, we will briefly examine how to model switchback experiments through the model mentioned above and how these assumptions may or may not hold in real applications.

2.3.1. Individual Level vs. Aggregate Level Modeling.

When it comes to switchback experiments, there are two approaches to consider: individual-level and aggregate-level modeling. The individual-level approach assumes that we can observe outcomes for each unit at every time period, which is ideal when the number of units is relatively small. This approach is common in clinical trials where patients are individually tracked as they receive both treatments and controls over time. In contrast, the aggregate-level approach is more practical when tracking individual units is not feasible. In the example of evaluating a new pricing policy on an online platform, it is impractical for the platform to monitor each customer individually due to the continuous influx of new customers daily and the overall large customer base. The more efficient and suitable way is to divide the

whole population into several groups (such as districts of a city in the ride-hailing platform) or simply take the revenue at each time period as the outcome of interest, which is a single number. Although we state our model in the context of individual level, one can just set the number of units $N = 1$ and get the corresponding result for the case of aggregate level.

2.3.2. Choice of the Surrogate Variable. In general, the central assumption in the model is Assumption 3, which states that the intermediate variable Q_{it} must be a surrogate for the past assignment. Finding such a surrogate may depend on the specific setting the experimenter is facing. In many applications, the surrogate variable can be chosen as a level of engagement within the system being analyzed. For instance, in the example of the ride-hailing platform, one could measure engagement individually by tracking how frequently each customer uses the service. Alternatively, from an aggregate modeling perspective, one could consider the total service demand at a given time period. Another example would be the crossover trial in the pharmaceutical industry, the intermediate variable in this case can be chosen as the related health metric for each patient at a specific time period. Although it relies on the experimenter's expertise to choose the intermediate variable, Assumption 3 is a condition on the observed outcomes, which means that we can perform statistical tests to determine whether this assumption holds in real applications. We briefly discuss how to perform statistical tests in the Online Appendix.

3. Estimation

After conducting the experiment, we observe a realized assignment path $\boldsymbol{w}_{1:T}$ and outcomes $\{Y_{it}, Q_{it}\}_{i\in[N],t\in[T]}$, with the goal of estimating the causal effect τ. In this section, we will discuss estimation without any restrictions on the form of the experiment as long as it satisfies the identification assumptions mentioned in the previous section. We first outline how the traditional IP score–based estimator can be applied to our setting. We then highlight that the IP estimator is not sample efficient, because it discards a significant portion of data near switching points. Leveraging the surrogacy structure mentioned in Section 2, we propose a novel importance sampling (IS)-based estimator that can utilize all the data points obtained after the experiment, leading to a more efficient estimation of the causal effect.

3.1. IP Score Estimator

The causal effect is often estimated using the propensity score (Horvitz and Thompson 1952). This approach can be directly applied to our setting because if we are in an experimental environment, the propensity score $\mathbb{P}(W_t|\boldsymbol{S})$ should be known to the experimenter. Here, we use $\boldsymbol{S} = (\boldsymbol{S}_1, \ldots, \boldsymbol{S}_T)$ to denote all the time covariates. Hence, one can construct the IP score estimator as follows:

$$\hat{\tau}^{IP} = \frac{1}{N(T-p)} \sum_{i=1}^{N} \sum_{t=p+1}^{T} Y_{it} \left(\frac{\mathbf{1}\{\boldsymbol{w}_{t-p:t} = \mathbf{1}_{p+1}\}}{\mathbb{P}(\boldsymbol{W}_{t-p:t} = \mathbf{1}_{p+1}|\boldsymbol{S})} - \frac{\mathbf{1}\{\boldsymbol{w}_{t-p:t} = \mathbf{0}_{p+1}\}}{\mathbb{P}(\boldsymbol{W}_{t-p:t} = \mathbf{0}_{p+1}|\boldsymbol{S})} \right). \quad (2)$$

The IP estimator (2) is widely used because it requires the least structure and information. In particular, it is unbiased if the treatment assignment and the potential outcomes are unconfounded.

Proposition 2. *Under Assumptions 1 and 2, the IP estimator is unbiased, i.e.,* $\mathbb{E}[\hat{\tau}^{IP}] = \tau$.

Although the IP estimator is unbiased with minimal assumptions, it is inefficient because it discards all data points near switching points, that is, those that do not satisfy $\boldsymbol{w}_{t-p:t} = \mathbf{1}_{p+1}$ or $\mathbf{0}_{p+1}$. For example, for a sample path of $\boldsymbol{w}_{1:5} = (1,1,0,0,1)$ with the order of carryover effect $p = 1$, the IP estimator will only use the data at time 2 and time 4 because they are the only time periods that a consecutive $p+1$ assignments are the same. Indeed, following the worst-case optimal experimental design proposed by Bojinov et al. (2023), only $1/p$ of the observed data points would be utilized in the worst case.

3.2. IS Estimator

The inefficiency associated with the IP estimator is generally unavoidable in the absence of structural information. However, given the surrogate structure mentioned in Section 2, we can utilize the presence of the intermediate variable to construct a more efficient estimator by using IS.

In Proposition 1, we decompose the estimand into two parts. The idea is to use importance sampling to reweight the intermediate variable and correct the impact of past assignments on the current outcome. In particular, if we assume the conditional density of the potential outcome $Q_{it}(\boldsymbol{w})$ given $\boldsymbol{S}_t$ and $\boldsymbol{X}_i$ for every assignment $\boldsymbol{w} \in \{0,1\}^p$ exists and is denoted as $f_{\boldsymbol{w}}(q|\boldsymbol{S}_t, \boldsymbol{X}_i)$. Then, we have the following relationship:

$$\begin{aligned}\mathbb{E}[Y_{it}(\mathbf{1}_{p+1})] &= \mathbb{E}_{Q\sim G(\mathbf{1}_p, \boldsymbol{S}_t, \boldsymbol{X}_i)}[h_t(Q, 1, \boldsymbol{S}_t, \boldsymbol{X}_i)] \\ &= \mathbb{E}_{Q\sim G(\boldsymbol{w}_{t-p:t-1}, \boldsymbol{S}_t, \boldsymbol{X}_i)} \left[\frac{f_{\mathbf{1}_p}(G)}{f_{\boldsymbol{w}_{t-p:t-1}}(G)} h_t(Q, 1, \boldsymbol{S}_t, \boldsymbol{X}_i) \right].\end{aligned}$$

Here, the ratio between the target density and the actual density of Q adjusts for the impact of past treatment assignments. In this way, if we assume the densities $f_{\boldsymbol{w}}$ are known to the experimenter, we can utilize

all the data points and obtain the following oracle estimator.

$$\hat{\tau}^{IS}_{oracle} := \frac{1}{N(T-p)} \sum_{i=1}^{N} \sum_{t=p+1}^{T} Y_{it} \left(\frac{\mathbf{1}\{W_t = 1\}}{\mathbb{P}(W_t = 1|\boldsymbol{S})} \cdot \frac{f_{\mathbf{1}_p}(Q_{it}|\boldsymbol{X}_i, \boldsymbol{S}_t)}{f_{\mathbf{W}_{t-p:t-1}}(Q_{it}|\boldsymbol{X}_i, \boldsymbol{S}_t)} - \frac{\mathbf{1}\{W_t = 0\}}{\mathbb{P}(W_t = 0|\boldsymbol{S})} \cdot \frac{f_{\mathbf{0}_p}(Q_{it}|\boldsymbol{X}_i, \boldsymbol{S}_t)}{f_{\mathbf{W}_{t-p:t-1}}(Q_{it}|\boldsymbol{X}_i, \boldsymbol{S}_t)} \right). \tag{3}$$

Proposition 3. *Under Assumption 1–3, the oracle estimator (3) is an unbiased estimator of the average treatment effect, that is,* $\mathbb{E}[\hat{\tau}^{IS}_{oracle}] = \tau$.

Proposition 3 gives us another identification strategy for the average treatment effect besides the IP estimator. In practice, the experimenter first estimates the conditional densities $f_w(q|\boldsymbol{x}, \boldsymbol{s})$ by $\widehat{f}_w(q|\boldsymbol{x}, \boldsymbol{s})$ after collecting the data and then substitutes them into Equation (3) to obtain a plug-in estimator $\hat{\tau}^{IS}$ as follows:

$$\hat{\tau}^{IS} := \frac{1}{N(T-p)} \sum_{i=1}^{N} \sum_{t=p+1}^{T} Y_{it} \left(\frac{\mathbf{1}\{W_t = 1\}}{\mathbb{P}(W_t = 1|\boldsymbol{S})} \cdot \frac{\widehat{f}_{\mathbf{1}_p}(Q_{it}|\boldsymbol{X}_i, \boldsymbol{S}_t)}{\widehat{f}_{\mathbf{W}_{t-p:t-1}}(Q_{it}|\boldsymbol{X}_i, \boldsymbol{S}_t)} - \frac{\mathbf{1}\{W_t = 0\}}{\mathbb{P}(W_t = 0|\boldsymbol{S})} \cdot \frac{\widehat{f}_{\mathbf{0}_p}(Q_{it}|\boldsymbol{X}_i, \boldsymbol{S}_t)}{\widehat{f}_{\mathbf{W}_{t-p:t-1}}(Q_{it}|\boldsymbol{X}_i, \boldsymbol{S}_t)} \right). \tag{4}$$

This approach incorporates all data points at each time period into the estimator through weighting, resulting in a potentially more efficient estimation strategy. The density estimation can be done by first aggregating all the observed intermediate variables Q_{it} with its corresponding lag-p treatment path $\boldsymbol{W}_{t-p:t-1}$, followed by applying standard methods for conditional density estimation, for example, conditional kernel density estimation (Hyndman et al. 1996), Generalized Additive Model for Location, Scale, and Shape (Rigby and Stasinopoulos 2005), conditional transformation models (Hothorn et al. 2014), and so on. This procedure is justified by Assumption 4 because Q_{it} is stationary across time.

4. Experimental Design

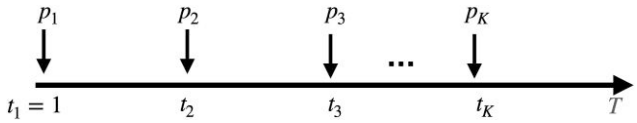
In this section, we focus on designing the experiment to ensure that the IS estimator achieves the minimum possible risk, specifically by minimizing the mean square error (MSE), $\mathbb{E}(\hat{\tau}^{IS} - \tau)^2$, between the estimated and true causal effects. For our theoretical analysis, we primarily examine the oracle estimator $\hat{\tau}^{IS}_{\text{oracle}}$, assuming the density of the intermediate variable is known. For the practical IS estimator, we provide a theoretical analysis for the case of a binary intermediate outcome in Section EC.2 of the Online Appendix. There, we show that the IS estimator with an estimated distribution achieves lower risk than its oracle counterpart—an observation common in semiparametric statistics (Hahn 1998, Henmi and Eguchi 2004, Hitomi et al. 2008). Additionally, in the next section, we present numerical experiments demonstrating that estimating the density does not reduce the efficiency of the estimator.

4.1. Switchback Experiments

Although there are various approaches to designing a switchback experiment, in this section, we focus on a family of switchback experiments introduced by Bojinov et al. (2023). This family of experiments is parametrized by a set of switching points $\mathbb{T} = \{1 = t_1 < t_2 < \cdots < t_K < t_{K+1} = T + 1\}$ and a corresponding set of probabilities $\mathbb{P} = (p_1, p_2, \ldots, p_K) \in (0,1)^K$. At each designated switching point t_i, the experimenter assigns treatment $(W_{t_i} = 1)$ with probability p_i or control $(W_{t_i} = 0)$ with probability $q_i = 1 - p_i$. During the interval $t \in (t_i, t_{i+1})$, the treatment or control remains unchanged from its assignment at t_i until the next switching point t_{i+1}. Figure 2 is an illustration of this family of experiments.

In this paper, we restrict our focus to a smaller class, in which the length of switching intervals is larger than the order of the carryover effect, that is, $t_{i+1} - t_i \geq p$ for $i = 1, 2, \ldots, K+1$. The reason for this is to facilitate our density estimation step in two ways. (1) By requiring each interval to be as least as long as p, we can collect sufficient sample points for the target distribution $f_{\mathbf{1}_p}$ and $f_{\mathbf{0}_p}$. (2) At the same time, it limits the number of distributions that need estimation. To illustrate this, denote R_p as the lag-p assignment paths that actually occur in the experiment. If the switching intervals are longer than p, the only possible patterns of p consecutive treatments are $R_p = \{\mathbf{0}_p, (1, \mathbf{0}_{p-1}), \ldots, (\mathbf{1}_{p-1}, 0), \mathbf{1}_p, (0, \mathbf{1}_{p-1}), \ldots, (\mathbf{0}_{p-1}, 1)\}$, which consists of $2p$ distributions. For other experimental designs with shorter intervals, the lag-p assignment may grow exponentially with p. In the worst case, there could be 2^p lag-p assignment patterns, making it impractical to estimate the densities for all patterns. We would like to point out that the optimal design proposed in Bojinov et al. (2023) also adheres to the principle that each interval should be longer than the order of carryover effects ($p + 1$ in our case), indicating that imposing this restriction should not significantly reduce the effectiveness of the experiment. For notational convenience, we

Figure 2. Set of Switchback Experiments That We Focus on



Note. Here, at each time t_i, the experimenter assigns treatment with probability p_i, and control with probability $1 - p_i$.

denote $\mathcal{T}_p$ to be the set of switching points $\mathbb{T}$ that satisfies $t_{i+1} - t_i \geq p$, that is,

$$\mathcal{T}_p = \{\mathbb{T} \mid t_{i+1} - t_i \geq p \text{ for } i = 1, 2, \ldots, K; K \in \mathbb{N}\}.$$

4.2. Feasible Domains

After defining the family of switchback experiments considered in this paper, we now define the feasible domains of the outcomes through two key definitions.

Definition 2. We say the potential outcomes $\{Y_{it}(\boldsymbol{w})\}_{\boldsymbol{w}\in\{0,1\}^{p+1}}$ are bounded by $A > 0$ if and only if $Y_{it}(\boldsymbol{w}) \in [0, A]$ for every $\boldsymbol{w} \in \{0,1\}^{p+1}$.

Definition 2 here defines the boundedness of the potential outcomes. For any potential outcomes that are bounded within region $[-A_0, A_1]$, one can add a constant $A_0 + A_1$ to the potential outcomes to ensure they are within region $[0, A_1 + A_0]$, and the treatment effect will not change. Although A here is a hyperparameter, we will demonstrate in the subsequent analysis that the magnitude of A does not influence the design of experiments. Essentially, we only require the existence of such a variable A, which is almost always the case in real applications. Next, we describe the influence of treatment assignments on the intermediate variable.

Definition 3. We say the conditional distribution of the potential outcomes $Q_{it}(\boldsymbol{w})$ are B-close if and only if

$$\mathbb{E}_{Q\sim f_{\boldsymbol{w}}(\cdot|\boldsymbol{x},\boldsymbol{s})}\left[\frac{f_{\boldsymbol{w}}(Q|\boldsymbol{x},\boldsymbol{s})}{f_{\boldsymbol{a}}(Q|\boldsymbol{x},\boldsymbol{s})}\right] \leq B, \quad \forall \boldsymbol{a} \in \{\mathbf{1}_p, \mathbf{0}_p\}, \boldsymbol{w} \in R_p,$$

for every $\boldsymbol{x} \in \mathbb{R}^d, \boldsymbol{s} \in \mathbb{R}^l$.

The parameter B quantifies the divergence between the distributions of the surrogate variable that appear around the switching points $(Q_{it}(\boldsymbol{w}), \boldsymbol{w} \in R_p)$ and the target distribution $Q_{it}(1_p)$ and $Q_{it}(0_p)$. This divergence reflects the impact of treatment assignments on the intermediate outcomes. The larger B is, the further the two sets of distributions can be, and $B = 1$ if and only if $Q_{it}(\boldsymbol{w})$ have the same distribution for each $\boldsymbol{w} \in R_p$.

To make the appropriate experimental design, the experimenter may need to specify the parameter B before the start of the experiment, which requires the experimenter's domain knowledge about the environment. If the company expects substantial change in the surrogate outcome Q_{it} under the new policy, B should be chosen to be larger. Conversely, if the new policy affects the intermediate variable Q_{it} only on a small scale, we can set B to be close to one. As we will show later, a larger B leads to longer switching intervals and a more conservative and robust way of designing the experiment, whereas a smaller B will lead to more frequent switches and potentially a more efficient estimator.

To provide intuition about the magnitude of B, consider two random variables Q_0 and Q_1 that follow normal distributions with the same standard error, that is, $Q_0 \sim \mathcal{N}(\mu_0, \sigma^2), Q_1 \sim \mathcal{N}(\mu_1, \sigma^2)$, then the distance measure $B = \mathbb{E}_{X\sim f_0}[f_0(X)/f_1(X)] = \mathbb{E}_{X\sim f_1}[f_1(X)/f_0(X)] = \exp((\mu_0 - \mu_1)^2/\sigma^2)$, where f_0 and f_1 are the densities for Q_0 and Q_1. Thus, two distributions being B-close can be approximately interpreted as their means satisfying $|\mu_0 - \mu_1| \leq \sigma \ln B$.

4.3. Minimax Framework

To derive the optimal experimental design, we adopt the minimax approach by minimizing the maximal risk of the oracle estimator based on the experimental design and the feasible domain of the potential outcomes. More specifically, the experimental design problem can be written as the following optimization problem:

$$\begin{aligned} \min_{\mathbb{T},\mathbb{P},K} \max_{Y_{it}(\boldsymbol{w}),Q_{it}(\boldsymbol{w})} \quad & \mathbb{E}[(\hat{\tau}^{IS}_{oracle} - \tau)^2] \\ \text{s.t.} \quad & \mathbb{T} \in \mathcal{T}_p \\ & \mathbb{P} \in (0,1)^K \\ & Y_{it}(\boldsymbol{w}) \text{ are bounded by } A \\ & Q_{it}(\boldsymbol{w}) \text{ are } B - \text{close}. \end{aligned} \tag{5}$$

This framework can also be seen in experimental design literature such as Basse et al. (2019) and Bojinov et al. (2023). We first prove that the inner maximization problem is solved optimally when the potential outcomes attain their maximum possible value and the distribution of the intermediate variable deviates as far as possible.

Proposition 4. *The solution to the inner maximization problem of* (5) *is* $Y_{it}(\boldsymbol{w}) \equiv A$, *almost surely and the conditional distributions* $f_{\boldsymbol{w}}$ *such that* $\mathbb{E}_{Q_{it}\sim f_{\boldsymbol{w}}(\cdot|\boldsymbol{x},\boldsymbol{s})}[f_{\boldsymbol{w}}(Q_{it}|\boldsymbol{x},\boldsymbol{s})/f_{\boldsymbol{a}}(Q_{it}|\boldsymbol{x},\boldsymbol{s})] = B$, $\forall \boldsymbol{a} \in \{\mathbf{1}_p, \mathbf{0}_p\}, \boldsymbol{w} \in R_p$.

Proposition 4 characterizes the worst case for the experimental problem. The intuition behind this is that if the potential outcomes take the maximal possible value, the variance introduced by the treatment assignments will also be maximized. This occurs because, in Inverse Propensity score Weighting-like estimators such as the IS estimator, the terms in the summation of (4) have similar absolute values, with the sign depending on the treatment status during that time period. Thus, a larger absolute value in the outcome will result in higher variances. A similar result is presented in Bojinov et al. (2023), which adopts a purely design-based perspective rather than a population-based perspective. Proposition 4 further enables us to transform the outer minimization problem into the following optimization problem.

Theorem 1. *The minimax problem* (5) *is equivalent to the following optimization problem:*

$$\min_{K,\mathbb{T},\mathbb{P}} \quad (t_2-p-1)^2\left(\frac{1}{p_1}+\frac{1}{q_1}\right)+\sum_{i=2}^{K}(t_{i+1}-t_i)^2\left(\frac{1}{p_i}+\frac{1}{q_i}\right) + p(B-1)\sum_{i=1}^{K-1}\left(\frac{p_i}{q_{i+1}}+\frac{q_i}{p_{i+1}}\right), \tag{6}$$

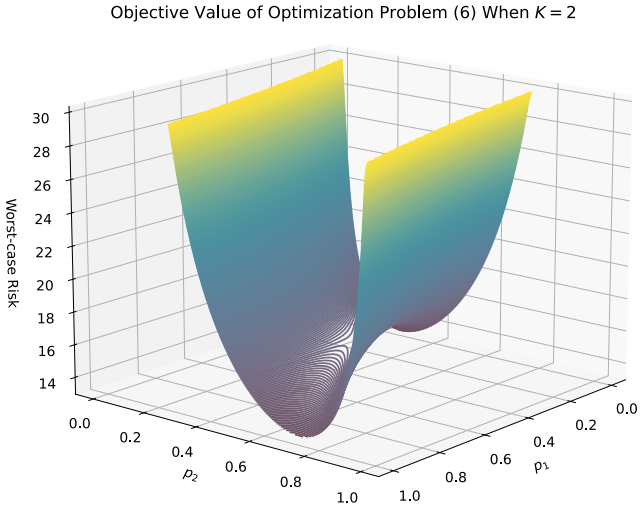
where we denote $q_i = 1-p_i$ *and* $t_{K+1} = T+1$.

In Equation (6), the parameter A does not appear, indicating that it does not influence the experimental design. The distance parameter B does appear in (6) in a way that it controls the tradeoff between the variance of the traditional estimation procedure, which is the first two terms, and the variance introduced by importance sampling near switching times, which is the last term. A larger B indicates greater variance introduced by the importance sampling step, as the two distributions differ more significantly.

Now we only need to solve the optimization problem (6) to get the worst-case optimal design of our importance sampling estimator $\hat{\tau}^{IS}_{oracle}$. Note that if $B=1$, which means the conditional distributions of $Q_{it}(\boldsymbol{w})$ are the same and thus carryover effect does not exist, the solution to the optimization problem (6) is to set $p_i = 1/2$ and $K = T$. This is the standard randomization scheme where at each time period, the experimenter assigns treatment or control with equal probability. However, for $B>1$, Problem (6) is generally nonconvex and hard to solve exactly. This is illustrated in Figure 3, where we set $p=1, K=T=2, t_2=3, t_3=4, B=6$ and plot the contour of the objective function. We can see directly that the objective function is not convex and has a saddle point at $(p_1,p_2)=(1/2,1/2)$.

Figure 3 also shows that the objective value at the saddle point is close to the optimal value. This observation motivates a heuristic solution, where we set $p_i = q_i = 1/2$ and seek the optimal design under this constraint. We present the design and prove it achieves almost asymptotically optimal performance while also having a nice nonasymptotic guarantee in the following section.

Figure 3. (Color online) Contour of the Objective Value of Optimization Problem (6) When $p=1, K=T=2, t_2=3, t_3=4, B=6$



Note. Under this setup, the optimization problem becomes $\max_{p_1,p_2} \frac{1}{p_1}+\frac{1}{1-p_1}+\frac{1}{p_2}+\frac{1}{1-p_2}+5\left(\frac{p_1}{1-p_2}+\frac{1-p_1}{p_2}\right)$.

4.4. Near-Optimal Design

Given that the optimization problem (6) is nonconvex and challenging to solve exactly, we propose a near-optimal experimental design in this section. To simplify the design and make it more practical, we assume that at each switching point, the experimenter assigns treatment or control with equal probability, that is, $p_i = q_i = 0.5$. Under this assumption, (6) reduces to

$$\min_{K,\mathbb{T}} \; 4\left[(t_2-p-1)^2+\sum_{i=2}^{K}(t_{i+1}-t_i)^2\right]+2p(K-1)(B-1). \tag{7}$$

For a fixed value of K, the Cauchy-Schwarz inequality implies that the minimum of (7) is achieved when the switching times $\{t_i\}_{i=1}^{K}$ are uniformly spaced, that is, $t_2-p-1 = t_3-t_2 = t_4-t_3 = \cdots = (T-p)/K$.[2] Substituting this condition, we can further simplify (7) as

$$\min_{K} \quad 4\frac{(T-p)^2}{K}+2p(K-1)(B-1).$$

The optimal number of switches is then approximately $K^* = \sqrt{2/(pB-p)}(T-p)$. Although this estimate disregards the requirement for integer values in intervals and switching times, it offers valuable insights for policy design. Based on the analysis above, we propose a heuristic policy design consisting of three components. First, we approximate the number of switches, denoted K^h, by using the derived expression for K^*. Next, we distribute the switching intervals uniformly over the time horizon. Finally, we assign equal probability for treatment and control in each interval, setting $p_i = q_i = 1/2$. The complete procedure is detailed in Algorithm 1.

In Algorithm 1, the switching time K^h includes an additional 0.5 in the denominator compared with the earlier proxy K^*. This additional term simplifies our theoretical analysis, and numerical experiments reveal negligible differences between these two approaches. The proposed policy suggests that the interval between switches is approximately $\max\{\sqrt{p(B-1)/2}, p\}$, an increasing function of both p and B. Extending each experimental period is reasonable when the order of the carryover effect is large, as it allows for sufficient data collection. Notably, when B is large, indicating

substantial variability in unit responses under different treatment paths, fewer switches with longer experimental periods are recommended to minimize variance in the importance sampling step at each switching point. Conversely, if the carryover effect is small (i.e., B is close to one), shorter experimental periods would be preferred as it could enhance efficiency. Next, we provide a theoretical guarantee that this heuristic design achieves near-optimal performance.

Algorithm 1 (Heuristic Experimental Design)

Input: Time horizon T, carryover effect length p, and distance measure B.

Output: The heuristic design $(K^h, \mathbb{P}^h, \mathbb{T}^h)$.

`/* Step 1: Set the number of switches. */`

if $\frac{\sqrt{2}(T-p)}{\sqrt{p(B-1)}} \geq T/p$ **then**

  Set $K^h = T/p$

else

  Set $K^h = k$ such that $k \in \mathbb{N}$ satisfies $\sqrt{k(k-1)} \leq \frac{\sqrt{2}(T-p)}{\sqrt{p(B-1)+0.5}} \leq \sqrt{k(k+1)}$.

end if

`/* Step 2: Distribute switching points evenly. */`

Suppose $T - p = a \cdot K^h + b$ where $a, b \in \mathbb{N}, 0 \leq b \leq K^h - 1$. Set

$$t_i = \begin{cases} a*(i-1)+2 & i = 2,3,\ldots,K^h - b + 1 \\ (a+1)*i + b - K^h - a + 1 & i = K^h - b + 2, \ldots, K^h + 1 \end{cases}$$

`/* Step 3: Set equal probabilities. */`

Set $p_i = q_i = 0.5$, $i \in [K^h]$.

Return: $(K^h, \mathbb{T}^h, \mathbb{P}^h)$.

Theorem 2. *Suppose the ratio between the objective value of (6) under the heuristic solution and the optimal solution is $R(T,B)$ for some fixed T and B. We have the following performance guarantee for our heuristic policy:*

1. (*Nonasymptotic*) *For any $T \geq 10$ and B satisfies $B \leq (T/p-1)^2 + 0.5$, we have $R(T,B) \leq 1.33$,*

2. *(Asymptotic) For a fixed B , suppose $T = mp$ for some $m \in \mathbb{N}$. Then we have the following upper bound on $R(T,B)$.*

- *If $1 \leq B \leq 2p+1$, then $\lim_{T\to\infty} R(T,B) = 1$.*
- *If $B > 2p+1$, then $\limsup_{T\to\infty} R(T,B) \leq \sqrt{1+\frac{1}{2p(B-1)}}$.*

Theorem 2 quantifies the performance gap between the heuristic policy and the optimal policy, showing that, in a nonasymptotic context, the variance of the proposed policy is at most 1.33 times that of the optimal policy. Notably, the assumption $B \leq (T/p-1)^2 + 0.5$ in the nonasymptotic setting is relatively mild, because in practical applications, the experimentation period T is typically large compared with the length of the carryover effect p. For instance, if the carryover effect is of order 1 and the intermediate variable Q_{it} follows a normal distribution with equal variance (i.e., $Q_{it}(0) \sim \mathcal{N}(\mu_1, \sigma^2)$ and $Q_{it}(1) \sim \mathcal{N}(\mu_2, \sigma^2)$), then B can be expressed as $B = \exp((\mu_1 - \mu_2)^2/\sigma^2)$. In this case, even with a small experimental period such as $T = 10$, the assumption holds as long as $\mu_1 - \mu_2 \leq 2.2\sigma$, which aligns with typical response variations in the application. This assumption only fails if there is a substantial response shift, which is uncommon in practice. Furthermore, our policy is asymptotically optimal when the bound B is small. When B is larger than $2p+1$, the upper bound of the performance gap is $\sqrt{1+1/(2pB-2p)} \leq \sqrt{1+1/(4p^2)}$, which remains close to one for integer values of $p \geq 1$.

4.5. Comparison with the IP Score Estimator

A natural question to ask is how much efficiency we can gain by first estimating the density and then doing the importance sampling. To end this section, we make a comparison between the performance of the oracle IS estimator $\hat{\tau}^{IS}_{oracle}$ and the IP score estimator $\hat{\tau}^{IP}$. Specifically, we calculate the worst-case variance of $\hat{\tau}^{IS}_{oracle}$ under the heuristic design in Algorithm 1 and compare it with the worst-case risk of $\hat{\tau}^{IP}$ under the optimal design proposed in Bojinov et al. (2023). The two estimators' worst-case risk ratio has the following bound.

Theorem 3. *Denote the worst-case risk of $\hat{\tau}^{IS}_{oracle}$ under the heuristic design in Algorithm 1 as r^{IS} and the worst-case risk of $\hat{\tau}^{IP}$ under the optimal design in Bojinov et al. (2023) as r^{IP}. Suppose $T = mp$ for some $m \in \mathbb{Z}^+$, then we have in the limiting case when $T \to \infty$, the ratio r^{IS}/r^{IP} has an upper bound*

$$\limsup_{T\to\infty} \frac{r^{IS}}{r^{IP}} \leq \begin{cases} \dfrac{2p+(B-1)}{8p} & \text{if } B < 2p+1 \\ \dfrac{\sqrt{2p(B-1)}}{4p} & \text{if } B \geq 2p+1. \end{cases}$$

Theorem 3 shows that as long as $B - 1 \leq 8p$, the estimator $\hat{\tau}^{IS}_{oracle}$ with the heuristic design is guaranteed to have a smaller risk. Again, For two normal distributions with the same standard deviation σ but different means μ_1 and μ_2, for their distance B to exceed $8p+1$, it must hold that $|\mu_1 - \mu_2| > d_p\sigma$, where $d_p = \sqrt{\log(8p+1)}$. We show this threshold with p ranging from one to five in Table 1.

From Table 1, we observe that even with a short carryover effect length ($p = 1$), the proposed estimator consistently achieves a smaller variance under the worst-case scenario, provided that the mean difference in responses satisfies $|\mu_1 - \mu_2| < 1.48\sigma$. As p increases,

Table 1. Threshold d_p for Different Length p of Carryover Effect

Length of carryover effect (p)	1	2	3	4	5
Threshold ($d_p = \sqrt{\log(8p+1)}$)	1.48	1.68	1.79	1.87	1.93

the threshold also rises due to the IP estimator potentially discarding more data, indicating that the IS estimator performs better across a broader range. In the context of ride-sharing platforms, where the outcome might represent revenue and the intermediate variable reflects driver activeness, it is unlikely for the mean activeness level to change by more than one standard deviation. Indeed, Xiong et al. (2023) demonstrate through an experiment conducted on a ride-sharing platform that the observed mean difference between the experimental outcome and the control outcome is on the order of 0.01σ.

Thus far, our analysis has focused on the oracle estimator, which assumes knowledge of the conditional density. For the IS estimator, which involves density estimation, direct theoretical analysis is challenging due to the variety of available estimation methods, each with distinct theoretical properties. To gain insights into the performance of the IS estimator, we conduct numerical experiments in Section 6. Surprisingly, these results reveal that the IS estimator often outperforms its oracle counterpart. Furthermore, in Section EC.2 of the Online Appendix, we provide a theoretical proof of this phenomenon for cases where the intermediate variable is binary. Based on both numerical and theoretical findings, we can confidently assert that with sufficient sample size, density estimation incurs little or no efficiency loss.

5. Inference and Testing

In this section, we discuss how to perform statistical inference based on the IS estimator (3). We first present how to do exact testing based on Fisher's randomization test. Then we demonstrate the IS estimator also enjoys a central limit theorem, and we propose an asymptotic way to construct a conservative confidence interval. Finally, we illustrate how to estimate the two central quantities in our experimental design—the order of carryover effect p and the distance measure B from past data or a pilot study.

5.1. Exact Inference: Fisher's Randomization Test

Exact inference can be performed using Fisher's randomization test (Fisher 1936, Rubin 1980), which tests the sharp null hypothesis through resampling and reconstructing the estimator. This method is applicable to any estimation strategy, including both the IP and IS estimators. For simplicity, we denote the generic estimator as $\hat{\tau}$ in this section. The sharp null hypothesis is given by

$$H_0 : (Y_{it}(\mathbf{1}_{p+1}), Q_{it}(\mathbf{1}_p)) = (Y_{it}(\boldsymbol{w}_{t-p:t-1}, w_t), Q_{it}(\boldsymbol{w}_{t-p:t-1})), \quad \forall \boldsymbol{w}_{t-p:t-1} \in \{0,1\}^p, w_t \in \{0,1\}. \tag{8}$$

This sharp null hypothesis represents that there is no treatment effect for any experimental unit at any time period for both the final outcome and the surrogate variable. Under the null, if we have observed one outcome $Y_{it}(1_{p+1})$, it can be seen as one sample from the other potential outcomes $Y_{it}(\boldsymbol{w})$ for $\boldsymbol{w} \in \{0,1\}^{p+1}$ as well. Thus, to perform the test, one can first resample the treatment assignment $\boldsymbol{W}_{1:T}$ according to the experimental mechanism and reconstruct the estimator based on the same observed outcomes. This yields a valid estimator under the null. The original estimator is then compared with the quantiles of the resampled distribution to compute the p-value. The procedure is detailed in Algorithm 2.

Algorithm 2 (Fisher's Exact Test for Sharp-Null Hypothesis Test)

Input: Total number of samples I, the estimated value $\hat{\tau}$.
Output: The p-value for the estimated value $\hat{\tau}$ under the sharp null (8).
for $i = 1 : I$ **do**
  Sample a new assignment path $\boldsymbol{w}_{1:T}^{[i]}$ according to the assignment mechanism.
  Hold the observed outcome Y_{it} and the intermediate variable Q_{it} unchanged, re-estimate the densities $\widehat{f}_{\boldsymbol{w}}$ and use the same method as $\hat{\tau}$ to construct the estimator $\hat{\tau}^{[i]}$
end for
Compute the p-value by $\hat{p} = I^{-1}\sum_{i=1}^{I} \mathbf{1}\{|\hat{\tau}^{[i]}| \geq |\hat{\tau}|\}$
Return: $\hat{p}$.

Confidence intervals for $\hat{\tau}$ can also be constructed by inverting the sequence of exact tests to identify regions where (8) is rejected at a prespecified nominal level. This way of doing test and inference is desirable in the sense that it is exact and also considers the variance when doing density estimation. However, it will require a lot of computation power, and the sharp-null hypothesis is usually a strong null hypothesis, leading to a reduced power. In the next section, we provide a way of doing asymptotic tests that require less computation but at a conservative level.

5.2. Asymptotic Tests

In this section, we discuss conducting asymptotic tests for the IS estimator. In particular, we adopt a limiting regime when N is fixed and $T \to \infty$ and derive a

central limit theorem for the oracle IS estimator (4). This regime is appropriate for switchback experiments because all the test units are assigned to the same policy, so the only possible regime for us to obtain the central limit theorem is by letting the length T go to infinity to have enough variation.

Theorem 4. *If we adopt the heuristic design in Algorithm 1, and suppose the estimator has nonnegligible variance, that is,* $\mathrm{Var}(\hat{\tau}^{IS}_{oracle}) \geq \Omega(T^{-1})$, *and the densities of the intermediate variable satisys* $\mathbb{E}_{Q_{it}\sim f_{\boldsymbol{w}}(\cdot|\boldsymbol{x},\boldsymbol{s})}\left[\frac{f^3_{\boldsymbol{w}}(Q_{it}|\boldsymbol{x},\boldsymbol{s})}{f^3_{\boldsymbol{a}}(Q_{it}|\boldsymbol{x},\boldsymbol{s})}\right] \leq U^3$, $\forall \boldsymbol{a} \in \{\mathbf{1}_p, \mathbf{0}_p\}$, $\boldsymbol{w} \in R_p$ *for some* $U > 0$. *Then under Assumptions 1–4, we have* $\hat{\tau}^{IS}_{oracle}$ *is asymptotically normal. Specifically,*

$$\frac{\hat{\tau}^{IS}_{oracle} - \tau}{\sqrt{\mathrm{Var}(\hat{\tau}^{IS}_{oracle})}} \xrightarrow{d} \mathcal{N}(0,1).$$

Theorem 4 serves as the foundation for us to perform asymptotic inference. The theorem has some finite population flavor in the sense that it requires a nonnegligible variance, which is also seen in Li and Ding (2017), Aronow and Samii (2017), and Bojinov and Shephard (2019). This is because we do not require the distribution of $Y_{it}(\boldsymbol{w})$ to be the same across t, leading to a nonstationary environment. The condition is satisfied if the variance does not originate from a few of the total time periods, which is true if the system does not go through a major shift during the experimental period. In Section EC.1 of the Online Appendix, we provide a method to estimate an upper bound of the variance $\mathrm{Var}(\hat{\tau}^{IS}_{oracle})$ from the observed data. Then after the variance $\hat{\sigma}$ is estimated, we can form an approximate asymptotic α-level confidence interval for the IS estimator by

$$\left[\hat{\tau}^{IS} - c_{\frac{1+\alpha}{2}}\hat{\sigma},\ \hat{\tau}^{IS} + c_{\frac{1+\alpha}{2}}\hat{\sigma}\right],$$

where $c_{\frac{1+\alpha}{2}}$ is the $\frac{1+\alpha}{2}$ quantile of the standard normal distribution. Although we derive the central limit theorem for the oracle estimator, we will show that it still enjoys adequate performance for the true estimator $\hat{\tau}^{IS}$ in Section 6.

5.3. Estimation of *B* and *p*

To implement the proposed experimental design, the experimenter needs to set the two hyper-parameters B and p. These parameters can either be determined based on the experimenter's domain expertise or estimated from data through a preliminary pilot study. Specifically, suppose the experimenter conducts another experiment to N_1 units and lasting for T_1 time periods with the assignment mechanism being randomly assigning treatment or control at each time period with equal probability. At the end of this pilot experiment, the experimenter obtains the data on the intermediate variable $\{Q_{it}\}_{i\in[N_1],t\in[T_1]}$.

To estimate the hyper-parameter B, the experimenter would initially have to estimate the conditional density of the intermediate variable Q_{it} from the data, which we denote by $\widehat{f}_{\boldsymbol{w}}(Q_{it}|X_i, S_t)$. Then, we can derive a plug-in estimator based on the definition of B in Definition 3 by

$$\hat{B} = \max_{\boldsymbol{w}\in R_p, \boldsymbol{a}\in\{1_p,0_p\}} \frac{1}{N_1T_1}\sum_{i=1}^{N_1}\sum_{t=1}^{T_1}\frac{\widehat{f}_{\boldsymbol{w}}(Q_{it}|\boldsymbol{X}_i,\boldsymbol{S}_t)}{\widehat{f}_{\boldsymbol{a}}(Q_{it}|\boldsymbol{X}_i,\boldsymbol{S}_t)}.$$

The estimation of the order of carryover effect p will depend on hypothesis testing. In particular, for a given $p_0 \in \mathbb{N}$, consider the following hypothesis testing problem:

$$H_0 : p < p_0 \leftrightarrow H_1 : p \geq p_0.$$

This is equivalent to testing whether the intermediate outcome Q_{it} is independent of the assignment preceding p_0 times, that is, W_{t-p_0}. Because the experimenter is assigning W_t independently, we can directly test whether the conditional distribution of $Q_{it}|W_{t-p_0} = 1$ and $Q_{it}|W_{t-p_0} = 0$ is the same by the two-sample Kolmogorov-Smirnov test. This is done by first aggregating the data $\mathcal{Q}_0 = \{Q_{it} : W_{t-p_0} = 0\}$ and $\mathcal{Q}_1 = \{Q_{it} : W_{t-p_0} = 1\}$ and then forming the empirical cumulative distribution function (CDF) F_0, F_1 for $\mathcal{Q}_0$ and $\mathcal{Q}_1$. The Kolmogorov-Smirnov statistics is the maximal difference between the two CDFs, that is, $D = \sup_q |F_0(q) - F_1(q)|$. The null hypothesis is rejected at level α if

$$D > \sqrt{-\ln\left(\frac{\alpha}{2}\right)\cdot\frac{|\mathcal{Q}_0| + |\mathcal{Q}_1|}{2|\mathcal{Q}_0|\cdot|\mathcal{Q}_1|}},$$

where $|\cdot|$ is the cardinality of a set. The experimenter can use the test for different p_0 and find the appropriate order of the carryover effect. Note that the pilot study only needs to be conducted once for us to perform the test at different levels of p_0.

6. Numerical Experiments

In this section, we numerically study the performance of the IS estimator and the proposed experimental design. In particular, there are three aspects that we will evaluate. (1) Prediction error: This includes comparisons across different estimators and designs and understanding how the error from density estimation will be propagated to the final estimator. (2) Inference power: We evaluate the power and coverage for the testing procedure and confidence interval in Section 5. (3) Robustness: We test the IS estimator under the scenario when the parameters p and B are misspecified. More numerical experiments, including on the impact of noise level, time horizon, number of units, and violation of the surrogacy assumption can be found in the Online Appendix.

As for the outcome model, we assume the carryover effect is of order 2 and the potential outcomes have a linear relationship with the treatment assignment. This model is widely adopted in the design of crossover trials in clinical studies (Hedayat and Afsarinejad 1978, Kenward and Jones 2014). In particular, the formulation is as follows:

$$\begin{aligned} Y_{it}(\boldsymbol{w}_{t-2:t})|Q_{it} &= \alpha_t + \delta_1 w_t + \delta_2 Q_{it} + (\boldsymbol{\alpha}_1^T \boldsymbol{x}_i + \boldsymbol{\alpha}_2^T \boldsymbol{s}_t) \\ &\quad + (\boldsymbol{\phi}_1^T \mathbf{x}_i + \boldsymbol{\phi}_2^T \boldsymbol{s}_t)\eta_{it}, \\ Q_{it}(\boldsymbol{w}_{t-2:t-1}) &= \gamma_1 w_{t-1} + \gamma_2 w_{t-2} + (\boldsymbol{\beta}_1^T \boldsymbol{x}_i + \boldsymbol{\beta}_2^T \boldsymbol{s}_t) \\ &\quad + (\boldsymbol{\theta}_1^T \boldsymbol{x}_i + \boldsymbol{\theta}_2^T \boldsymbol{s}_t)\epsilon_{it}, \end{aligned} \tag{9}$$

where α_t is the (unknown) fixed effect associated with time t; $\delta_1, \delta_2, \gamma_1$, and γ_2 are nonstochastic coefficients; $\boldsymbol{x}_i$ and $\boldsymbol{s}_t$ are the covariates for individual i and time t; $\boldsymbol{\alpha}_j, \boldsymbol{\beta}_j, \boldsymbol{\phi}_j, \boldsymbol{\theta}_j$ for $j \in \{1,2\}$ are the coefficient for the covariates; and η_{it} and ϵ_{it} are two independent and identically distributed random noises and are assumed to follow the standard normal distribution.

We would like to make a few comments on Model (9). (1) The model satisfies the surrogacy assumption (Assumption 3) because the potential outcome $Y_{it}(\boldsymbol{w}_{t-2:t})$ is directly modeled through the intermediate variable Q_{it}. Conditioning on Q_{it}, the past assignments w_{t-2}, w_{t-1} no longer appear in the first equation of (9). (2) It also satisfies the stationarity assumption for the intermediate variable Q_{it} (Assumption 4), which could be seen by the construction of Q_{it}. (3) The model permits unmeasured nonstationarity in the final outcome Y_{it} through the drift term α_t, which introduces nonstationarity even after conditioning on the covariates $\boldsymbol{x}_i$ and $\boldsymbol{s}_t$. (4) The model allows heteroskedasticity because the noise terms are associated with the covariates.

Under this setting, the treatment effect is given by $\tau = \delta_1 + \delta_2(\gamma_1 + \gamma_2)$. There are four possible combinations of (w_{t-2}, w_{t-1}), which lead to four different distributions of the potential outcomes Q_{it}. For a particular $(w_{t-2}, w_{t-1}) \in \{0,1\}^2$, the distribution of $Q_{it}(w_{t-2}, w_{t-1})$ is a normal distribution with mean $\gamma_1 w_{t-1} + \gamma_2 w_{t-2} + (\boldsymbol{\beta}_1^T \boldsymbol{x}_i + \boldsymbol{\beta}_2^T \boldsymbol{s}_t)$ and variance $(\boldsymbol{\theta}_1^T \boldsymbol{x}_i + \boldsymbol{\theta}_2^T \boldsymbol{s}_t)^2$. The distance measure can be calculated as $B = \max_{i,t} \exp((\gamma_1 + \gamma_2)^2 / (\boldsymbol{\theta}_1^T \boldsymbol{x}_i + \boldsymbol{\theta}_2^T \boldsymbol{s}_t)^2)$.[3]

6.1. Prediction Error

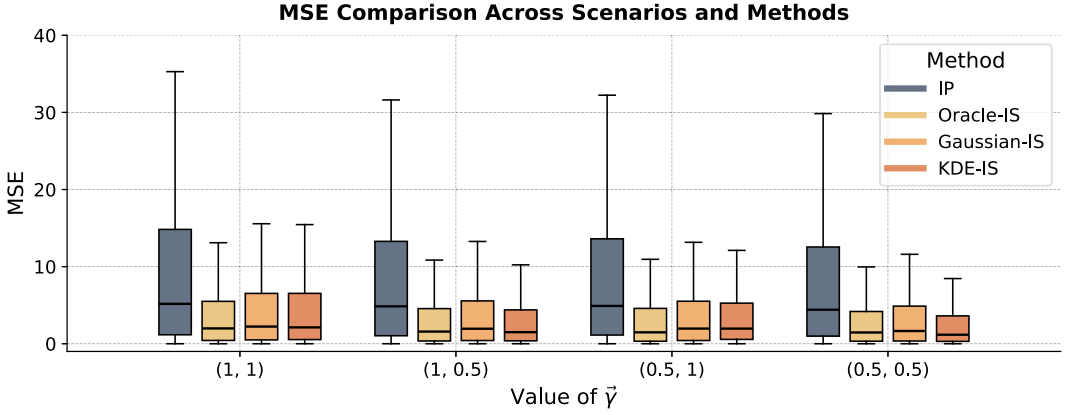
We begin by testing the prediction error of the estimator $\hat{\tau}^{IS}$. Our experiment focuses on two key questions. (1) To what extent does the importance sampling step improve accuracy? (2) How does the proposed experimental design help reduce risk? To address these questions, we set $\delta_1 = \delta_2 = \sigma_1 = \sigma_2 = 1$ and allow γ_1 and γ_2 to take different values. This setup enables us to evaluate the performance of the estimator and experimental design under varying treatment effects τ and distance measures B.

For the IS estimator, we test three variants based on the density estimation method. Oracle-IS: The IS estimator that uses the true density of the intermediate variable. Gaussian-IS: The joint density of $(Q_{it}(\boldsymbol{w}_{t-p:t-1}), \boldsymbol{X}_i, \boldsymbol{S}_t)$ is modeled as a Gaussian distribution, and the parameters are estimated via maximum likelihood. KDE-IS: The conditional density is obtained using nonparametric kernel density estimation (KDE).

We compare the MSE of the IP estimator and the IS estimator across four different setups. For $\hat{\tau}^{IP}$, we adopt the design proposed by Bojinov et al. (2023), where treatment and control switches occur at time periods $\mathbb{T}_1 = \{1, 7, 10, 13, \ldots, T-12, T-9, T-6, T\}$ with equal probabilities. For $\hat{\tau}^{IS}$, we use the proposed policy described in Section 4. The results are presented in Figure 4.

As shown in Figure 4, all three variants of the IS estimator achieve significantly lower MSE than the IP estimator, reducing risk by nearly 50%. Additionally, the MSEs of the Gaussian-IS and KDE-IS are relatively similar, indicating that the density estimation step does not substantially influence the performance of

Figure 4. (Color online) Mean Squared Error of the IP Estimator and Three Variants of the IS Estimator



Note. The boxplot includes the 25th, 50th, and 75th percentile of the MSE across 1,000 experiments.

the IS estimator. Interestingly, the IS estimator with estimated densities occasionally outperforms its oracle counterpart, a phenomenon commonly observed in semiparametric statistics (Hahn 1998, Henmi and Eguchi 2004). We discuss this phenomenon theoretically in Section EC.2 of the Online Appendix.

Next, we examine the impact of the experimental design. In particular, we compare the IS estimator under the proposed design and two heuristic designs where the switches are made every p or $p+1$ time periods, that is, $\mathbb{T}=\{1,3,5,\dots,T\}$ or $\{1,4,7,\dots,T\}$. Note that for the design of the IP estimator, Bojinov et al. (2023) requires the interval length to be at least $p+1=3$. In contrast, leveraging the surrogacy structure allows us to make shorter intervals ($\geq p$), enabling more frequent switches. We construct the IS estimator under three different estimation methods and present the results in Figure 5.

From Figure 5, we observe that the IS estimator under the proposed design achieves a lower risk compared with the heuristic designs, underscoring the effectiveness of the proposed method. Moreover, this advantage holds across all estimation methods, indicating that experimental design plays a crucial role in reducing the risk of the estimator.

6.2. Inference Power

In this section, we illustrate the effectiveness of the inference procedure proposed in Section 5. We perform both exact and asymptotic inference procedures, addressing two key questions. (1) What is the power of the inference procedures? Specifically, given that the treatment effect is not zero, what are their abilities to reject the null hypothesis? (2) What is the true coverage probability of the confidence interval?

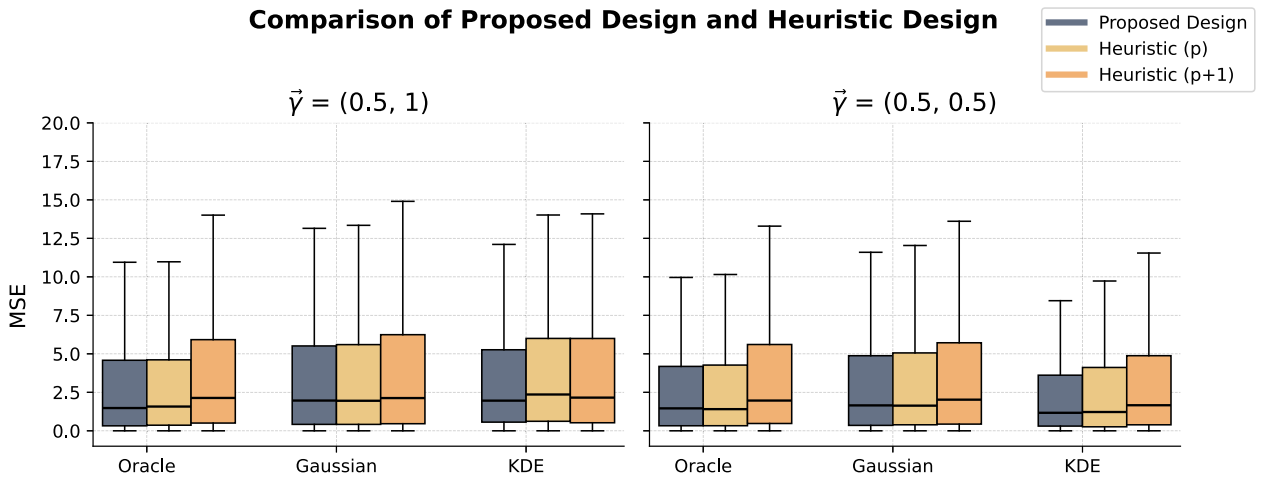
For this analysis, we use the linear model (9) and set $\delta_2=\sigma_1=\sigma_2=1$ with $(\gamma_1,\gamma_2)=(0.5,0.5)$. We vary $\delta_1\in\{0,1,2,3\}$ to simulate treatment effects of different magnitudes, corresponding to average treatment effects $\tau=1,2,3,4$. The estimator used is the Gaussian-IS estimator. The hypothesis being tested is the sharp null hypothesis (8) for exact inference and the regular null hypothesis $H_0:\tau=0\leftrightarrow H_1:\tau\neq 0$ for asymptotic inference. The significance level is set at 0.05. Additionally, we report the probability that the asymptotic conservative 95% confidence interval, constructed as described in Section 5.2, covers the true treatment effect τ. The results are presented in Figure 6.

From Figure 6, (a) and (b), we observe that both inference procedures successfully reject the false null hypothesis, with rejection probabilities increasing as the time periods lengthen and the treatment effect becomes more pronounced. This is because a longer experimental period provides more data, and a larger treatment effect strengthens the signal relative to noise, facilitating more robust inference. Comparing Figure 6, (a) and (b), we observe no significant difference between the two methods, indicating that the asymptotic confidence interval closely approximates the true confidence interval. This conclusion is further supported by Figure 6(c), which shows that the true coverage probability is consistently around or above the designated 95% level. This result highlights the conservative nature of the proposed variance estimator.

6.3. Misspecification of *B* and *p*

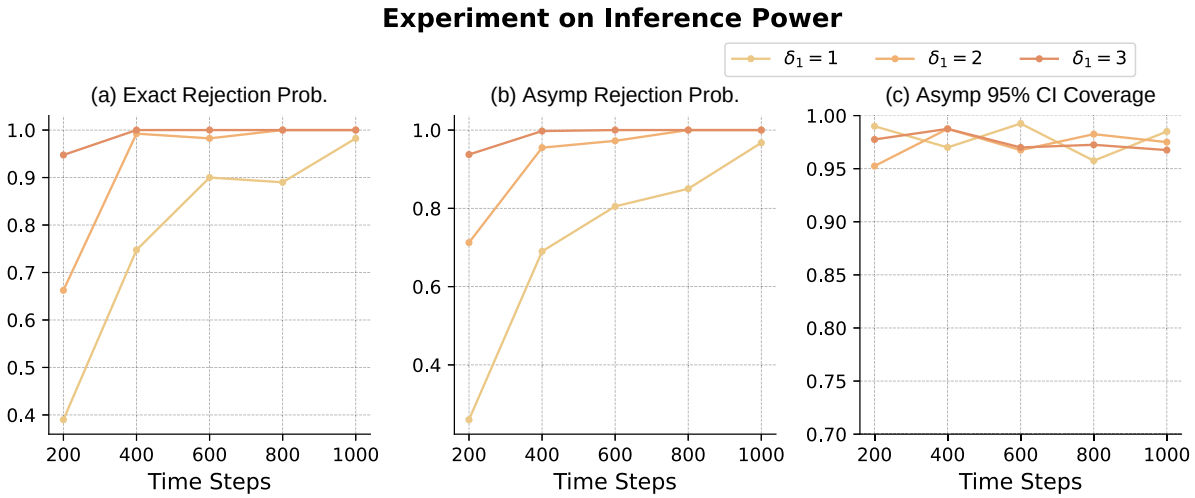
Finally, we examine the robustness of our model under misspecification of the parameters B and p. First, we consider the scenario where B is incorrectly estimated, and compare the MSE of our estimator with three baselines: IP, IS-H (p), and IS-H ($p+1$). The notation IS-H (p) indicates an IS estimator that uses a heuristic switching design every p time steps, and IS-H ($p+1$) applies a switching interval of length $p+1$.

Figure 5. (Color online) Comparison of MSE of the Proposed Design and Two Other Heuristic Designs



Note. Heuristic (p)/($p+1$) denotes the design that makes switches every p or $p+1$ time periods.

Figure 6. (Color online) Coverage Probability of the Confidence Interval and Rejection Rate Under True Alternative Hypothesis



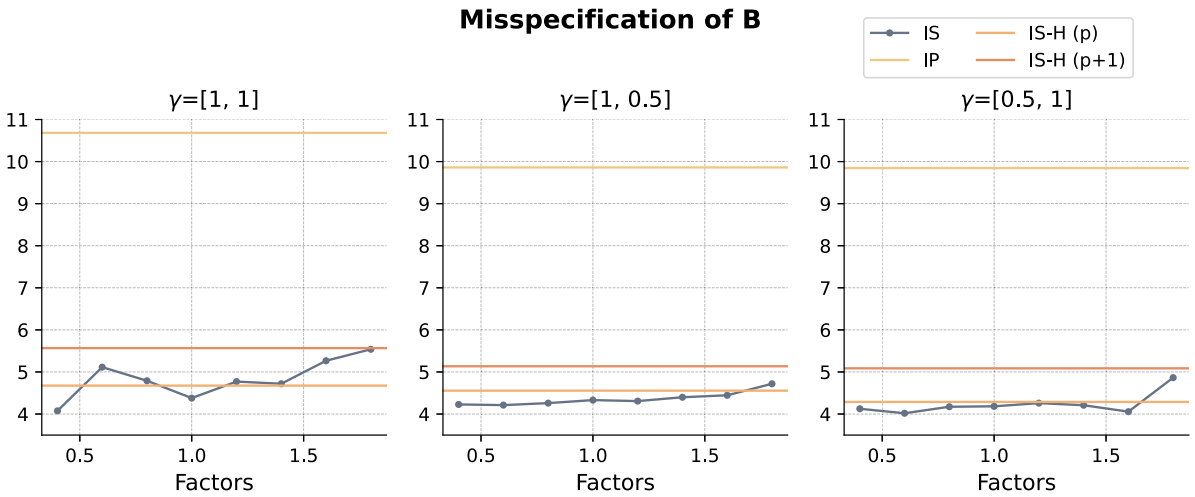
Both IS and IS-H use maximum likelihood estimation (MLE) under a joint Gaussian model for density estimation. We vary the true B by multiplying it by a factor ranging from 0.5 to 2 to assess the MSE of the four estimators in three different γ-scenarios. The results are presented in Figure 7.

Figure 7 shows that the proposed IS estimator remains robust across a wide range of B values, consistently exhibiting lower MSE than the heuristic counterparts and the IP estimator. When $\gamma = (1,1)$, the treatment effect is large, leading to a larger B and fewer switches. Under these conditions, heuristic approaches are less effective, whereas the proposed method identifies an experimental design that attains a lower risk. In the other two scenarios, where B is relatively small, differences in MSE across different values of B are less pronounced.

Next, we investigate the impact of misspecifying p. Such misspecification can affect the four estimators differently. For the IP and IS-H $(p/p+1)$ estimators, the parameter p influences both the experimental procedure and the estimation process. In contrast, for the IS estimator, the experimental design is independent of p, so errors in p primarily affect estimation rather than design, potentially enhancing its robustness against different p. Here, we assume that a small sample of experimental data are available to estimate B using the misspecified p. Although p may be incorrectly specified in estimating B, the process remains consistent as long as the specified p is greater than the true value. The MSE of the three estimators across different scenarios is shown in Figure 8.

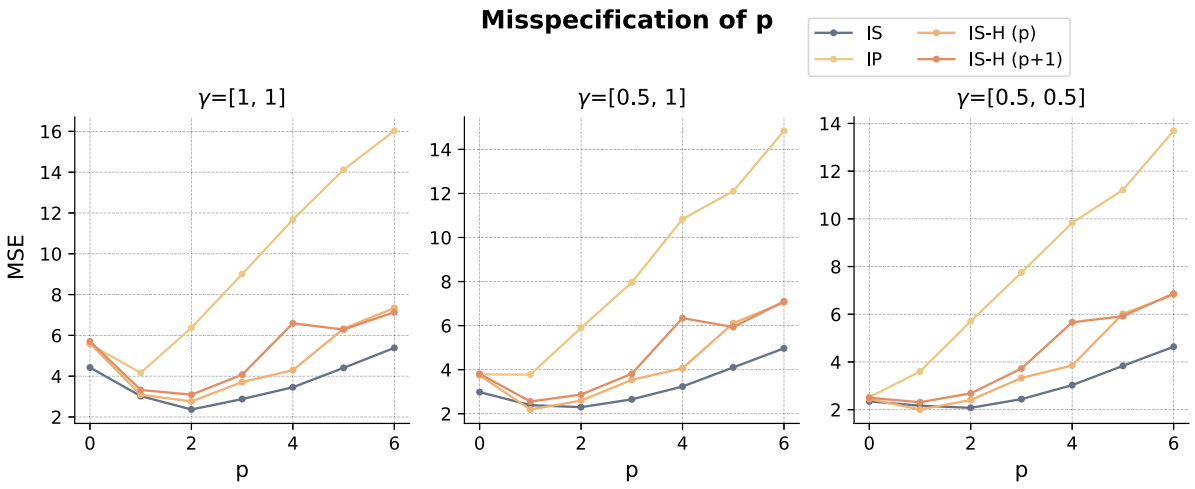
From Figure 8, we see that the IS estimator is relatively robust to errors in p. When the specified p is smaller than the actual value of two, all estimators exhibit large MSE due to inconsistency introduced by the misspecification. However, as p increases beyond

Figure 7. (Color online) MSE When the Distance Measure B Is Mis-Specified



Note. The B used in the algorithm is the actual B multiplied by the factor on the x axis.

Figure 8. (Color online) MSE When the Length of the Carryover Effect p Is Mis-Specified



Note. The true value of p is two under all three cases.

the true value, the MSE of the IP and IS-H estimators increases again because their experimental designs are adversely influenced by the incorrect p, resulting in fewer switches. In contrast, because the proposed IS design does not depend on p, its MSE remains comparatively low, underscoring its robustness under parameter misspecification.

7. Conclusion

In this paper, we consider the problem of conducting switchback experiments in a setting where there exists a surrogate variable. This surrogate variable is assumed to block the causal path from the past assignment to current outcomes, effectively capturing the potential temporal interference. Building on this structure, we propose an estimator based on importance sampling, which applies a likelihood ratio weighted adjustment to data points near switching times. Then we discuss how to best conduct the switchback experiments if the proposed estimator were being used. We devise a near-optimal heuristic policy that minimizes the worst-case risk and show that the heuristic policy has a nice performance guarantee in both the asymptotic and nonasymptotic settings. Both exact inference and asymptotic inference procedures are established based on the estimator, which facilitates the estimator's statistical application. We demonstrate by numerical experiments that the proposed estimator along with the proposed design has a smaller risk compared with the current approach in estimating the treatment effect.

Overall, our estimation method and experimental design offer companies an alternative approach for conducting switchback experiments. The key insight is that companies should evaluate whether specific causal structures, such as the surrogacy structure discussed here, are present before conducting experiments. Identifying such structures can substantially enhance the efficiency of the experimental process.

Endnotes

[1] Note that the i.i.d. assumption here does not imply there is no interference between the units because we do not characterize the scenario where units receive different treatment assignments.

[2] We disregard the fact that $(T-p)/K$ may not be an integer.

[3] In the experiment, instead of using the maximum, which appears to be over conservative, we recommend using the mean for the construction of B. That is, we take $B = \frac{1}{NT}\sum_i\sum_t \exp((\gamma_1+\gamma_2)^2/(\boldsymbol{\theta}_1^T\boldsymbol{x}_i + \boldsymbol{\theta}_2^T\boldsymbol{s}_t)^2)$.

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